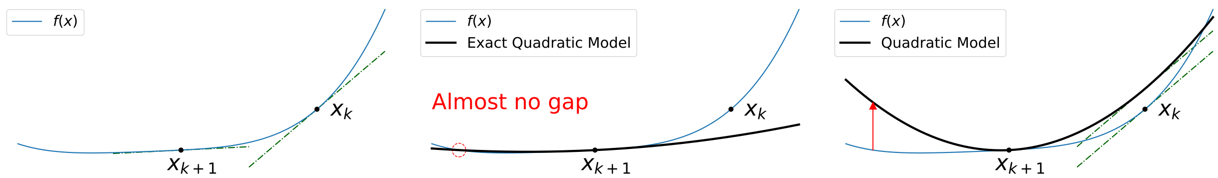

Figure 1: Limitations of the standard secant condition. (a) Function values and gradients at x_k and x_{k+1} are already known. (b) The ideal quadratic model constructed from the exact Hessian fits well around the new point x_{k+1}. (c) The standard secant condition may fail to capture curvature adequately.

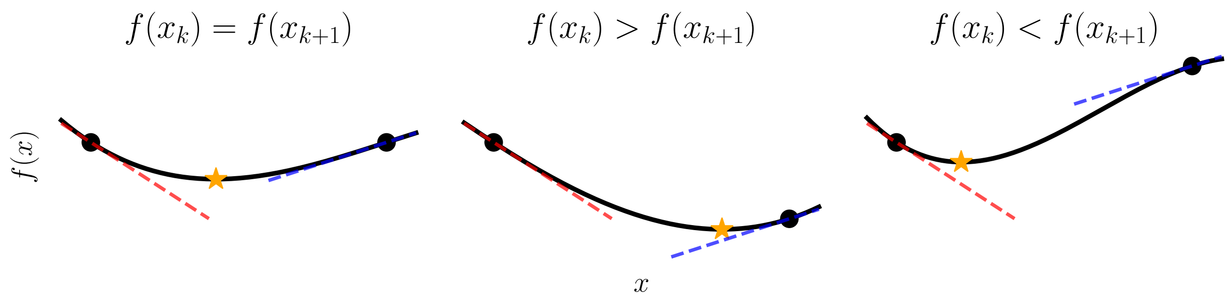

Figure 2: Basic idea of the modified secant condition. Even when the gradients are the same at two points x_k and x_{k+1}, natural interpolation of function values differs depending on the function values $f(x_k)$ and $f(x_{k+1})$.

1 Modified Secant Condition

This section explains the modified secant condition, a modification of the standard secant condition. The modified secant condition incorporates function value information in addition to gradient information, leading to more accurate Hessian approximations.

1.1 Derivation of the Modified Secant Condition

Although the secant condition is widely used in many quasi-Newton methods, it can fail to capture the curvature of the objective function accurately because it only utilizes gradient information. We illustrate this limitation in Figure 1. In Figure 1, we have two points x_k and x_{k+1} where function values and gradients are already known. We see that the ideal quadratic model constructed from the exact Hessian fits well around the new point x_{k+1}, leading to a better convergence behavior. However, the standard secant condition uses only gradient information and ignores function values. Consequently, it may significantly misestimate curvature, leading to a poor approximation of the true objective function.

We can overcome this limitation by utilizing the function values. The basic idea is illustrated in Figure 2. Even when the gradients are the same at two points x_k and x_{k+1}, natural interpolation of function values differs depending on the function values $f(x_k)$ and $f(x_{k+1})$. This observation motivates the modified secant condition, which incorporates function value information to improve Hessian approximation.

In the following, we present two well-known modified secant conditions.

1.1.1 Function-Value-Matching Modified Secant Condition

The first modification enforces the quadratic model to match the function value at the previous point [1, 4, 6]. Let us consider an alternate model with a different approximate Hessian B^{F}_{k+1}:

$$m^{\mathrm{F}}_{k+1}(x) := f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B^{\mathrm{F}}_{k+1}(x - x_{k+1}).$$

We require this model to satisfy

$$m^{\mathrm{F}}_{k+1}(x_k) = f(x_k),$$

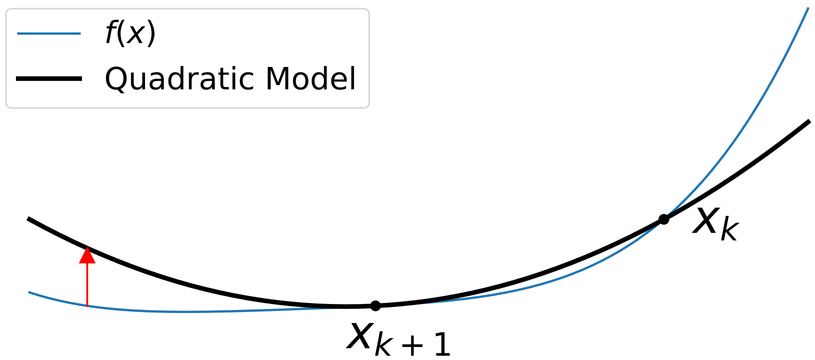

Figure 3: Function-value-matching modified secant equation. The function values $f(x_k)$ and $m_k^{\mathrm{F}}(x_k)$ are matched at the previous point x_k.

which enforces that the function value at the previous point is correctly modeled.

Let us derive the corresponding modified secant condition for B_{k+1}^{F}. Substituting the definition of m_{k+1}^{F} and using $s_k = x_{k+1} - x_k$, the condition becomes

$$
\begin{aligned}
f(x_k) &= f(x_{k+1}) + \nabla f(x_{k+1})^\top (x_k - x_{k+1}) + \frac{1}{2}(x_k - x_{k+1})^\top B_{k+1}^{\mathrm{F}} (x_k - x_{k+1}) \\
&= f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2} s_k^\top B_{k+1}^{\mathrm{F}} s_k.
\end{aligned}
$$

Now, instead of the standard secant condition, we assume a modified form with an additional term involving the identity matrix:

$$
\left(B_{k+1}^{\mathrm{F}} + \sigma_k^{\mathrm{F}} I\right) s_k = y_k.
$$

Here, $\sigma_k^{\mathrm{F}} \in \mathbb{R}$ is a scalar to be determined as follows. Substituting this equation yields

$$
f(x_k) = f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2} s_k^\top y_k - \frac{\sigma_k^{\mathrm{F}}}{2} \|s_k\|^2 .
$$

Solving for σ_k^{F} and using $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, we obtain

$$
\begin{aligned}
\sigma_k^{\mathrm{F}} &= \frac{2(f(x_{k+1}) - f(x_k)) - (2\nabla f(x_{k+1}) - y_k)^\top s_k}{\|s_k\|^2} \\
&= \frac{2(f(x_{k+1}) - f(x_k)) - (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2}.
\end{aligned}
$$

Therefore, the modified secant condition becomes

$$
B_{k+1}^{\mathrm{F}} s_k = \hat{y}_k^{\mathrm{F}} \coloneqq y_k + \frac{2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2} s_k.
$$

By using this modified $\hat{y}_k^{\mathrm{F}}$ instead of y_k in various quasi-Newton updates, we can incorporate function value information while keeping almost the same algorithm. Additionally, since condition $s_k^\top y_k > 0$ is important for maintaining positive definiteness in BFGS update, we can consider the following modification to ensure that the inner product with s_k remains positive:

$$
B_{k+1}^{\mathrm{F}'} s_k = y_k + \frac{\max(0, 2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k)}{\|s_k\|^2} s_k.
$$

This is the function-value-matching modified secant condition. Refer to Figure 3 for an illustration.

1.1.2 Cubic-Augmented Modified Secant Condition

The second modification secant condition introduces a cubic term to the model, allowing simultaneous satisfaction of both the function value and the gradient matching at the previous point [5, 7, 8]. Let $T_{k+1} \in \mathbb{R}^{n\times n\times n}$ be the third-order derivative tensor of f at x_{k+1} such that

$$s_k^\top (T_{k+1}s_k)s_k = \sum_{i,j,l=1}^{n} \partial_{ijl} f(x_{k+1}) s_k^{(i)} s_k^{(j)} s_k^{(l)}.$$

Here, $\partial_{ijl} f$ denotes the third derivative of f with respect to x_i, x_j, and x_l, and $s_k^{(i)}$ is the i-th component of the vector s_k. We introduced this tensor solely for analysis purposes, and it will be eliminated from the final formula.

By incorporating this tensor term, we can define the following model:

$$\begin{aligned} m_{k+1}^{\mathrm{C}}(x) &:= f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B_{k+1}^{\mathrm{C}}(x - x_{k+1}) \\ &\quad + \frac{1}{6}(x - x_{k+1})^\top (T_{k+1}(x - x_{k+1}))(x - x_{k+1}). \end{aligned}$$

With this model, we can enforce both function value and gradient matching at the previous point x_k. Specifically, we require

$$\begin{cases} m_{k+1}^{\mathrm{C}}(x_k) = f(x_k), \\ \nabla m_{k+1}^{\mathrm{C}}(x_k) = \nabla f(x_k). \end{cases}$$

By substituting the definition of m_{k+1}^{C} and using $s_k = x_{k+1} - x_k$, we can rewrite these conditions as

$$\begin{aligned} f(x_k) &= f(x_{k+1}) - s_k^\top \nabla f(x_{k+1}) + \frac{1}{2} s_k^\top B_{k+1}^{\mathrm{C}} s_k - \frac{1}{6} s_k^\top (T_{k+1}s_k)s_k, \\ \nabla f(x_k) &= \nabla f(x_{k+1}) - B_{k+1}^{\mathrm{C}} s_k + \frac{1}{2}(T_{k+1}s_k)s_k. \end{aligned}$$

Rearranging and multiplying both sides by 3 for the first equation, and taking the inner product with s_k for the second equation, we have

$$\begin{aligned} 3(f(x_k) - f(x_{k+1})) + 3s_k^\top \nabla f(x_{k+1}) &= \frac{3}{2} s_k^\top B_{k+1}^{\mathrm{C}} s_k - \frac{1}{2} s_k^\top (T_{k+1}s_k)s_k, \\ -s_k^\top y_k &= -s_k^\top B_{k+1}^{\mathrm{C}} s_k + \frac{1}{2} s_k^\top (T_{k+1}s_k)s_k. \end{aligned}$$

By adding these and using $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, we eliminate the tensor term and obtain the scalar identity

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2} s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) + \frac{1}{2} s_k^\top y_k = \frac{1}{2} s_k^\top B_{k+1}^{\mathrm{C}} s_k.$$

Again, for some scalar $\sigma_k^{\mathrm{C}} \in \mathbb{R}$, we assume $\left(B_{k+1}^{\mathrm{C}} + \sigma_k^{\mathrm{C}} I\right) s_k = y_k$. Then, the previous equation yields

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2} s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) = -\frac{\sigma_k^{\mathrm{C}}}{2} \|s_k\|^2,$$

which means

$$\sigma_k^{\mathrm{C}} = -\frac{6(f(x_k) - f(x_{k+1})) + 3s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2}.$$

Thus, the modified secant condition becomes

$$B_{k+1}^{\mathrm{C}} s_k = y_k + \frac{6(f(x_k) - f(x_{k+1})) + 3s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2} s_k.$$

This is a cubic-augmented modified secant condition. Refer to Figure 4 for an illustration.

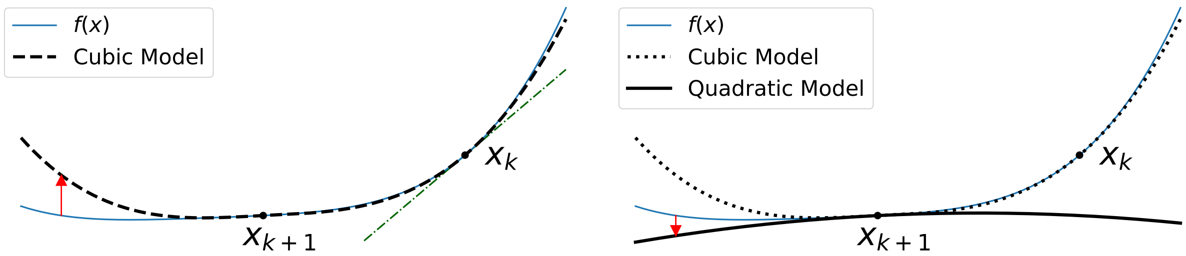

Figure 4: Cubic-augmented modified secant equation. The cubic term is incorporated into the model to simultaneously match both function value and gradient at the previous point. The model is constructed using the expansion up to the quadratic term.

1.2 Other Curvature Related Methods

There are several other topics related to obtaining the curvature pairs s_k, y_k. Agg-BFGS [2] is an alternative approach to managing curvature information by aggregating data rather than discarding the oldest information and adding the newest. Multi-Secant [3] extends the secant condition framework by maintaining multiple pairs of step and gradient difference vectors. In standard form we have seen, it is defined as

$$s_i = x_{i+1} - x_i, \quad y_i = \nabla f(x_{i+1}) - \nabla f(x_i). \quad (i = k-m, \ldots, k)$$

In contrast, Multi-Secant centers all vectors around the latest iterate:

$$s_i = x_{k+1} - x_i, \quad y_i = \nabla f(x_{k+1}) - \nabla f(x_i). \quad (i = k-m, \ldots, k)$$

This approach can lead to better approximations in some cases.

References

[1] Saman Babaie-Kafaki. A modified BFGS algorithm based on a hybrid secant equation. *Science China Mathematics*, 54(9):2019–2036, 2011. ISSN 1869-1862. doi: 10.1007/s11425-011-4232-7.

[2] Albert S. Berahas, Frank E. Curtis, and Baoyu Zhou. Limited-memory BFGS with displacement aggregation. *Mathematical Programming*, 194(1):121–157, 2022. ISSN 1436-4646. doi: 10.1007/s10107-021-01621-6.

[3] Mokhwa Lee and Yifan Sun. Advancing Multi-Secant Quasi-Newton Methods for General Convex Functions, 2025.

[4] Zengxin Wei, Guoyin Li, and Liqun Qi. New quasi-Newton methods for unconstrained optimization problems. *Applied Mathematics and Computation*, 175(2):1156–1188, 2006. ISSN 0096-3003. doi: 10.1016/j.amc.2005.08.027.

[5] Hiroshi Yabe, Hideho Ogasawara, and Masayuki Yoshino. Local and superlinear convergence of quasi-Newton methods based on modified secant conditions. *Journal of Computational and Applied Mathematics*, 205(1):617–632, 2007. ISSN 03770427. doi: 10.1016/j.cam.2006.05.018.

[6] Ya-Xiang Yuan. A Modified BFGS Algorithm for Unconstrained Optimization. *IMA Journal of Numerical Analysis*, 11(3):325–332, 1991. ISSN 0272-4979. doi: 10.1093/imanum/11.3.325.

[7] J. Z. Zhang, N. Y. Deng, and L. H. Chen. New Quasi-Newton Equation and Related Methods for Unconstrained Optimization. *Journal of Optimization Theory and Applications*, 102(1):147–167, 1999. ISSN 0022-3239, 1573-2878. doi: 10.1023/A:1021898630001.

[8] Jianzhong Zhang and Chengxian Xu. Properties and numerical performance of quasi-Newton methods with modified quasi-Newton equations. *Journal of Computational and Applied Mathematics*, 137(2):269–278, 2001. ISSN 0377-0427. doi: 10.1016/S0377-0427(00)00713-5.